

C03 AT2 – MEDIUM LEVEL CLASS TEST

Q1 Default and Parameterized Constructors

20 marks

Write a C++ program that defines a class Book with attributes title, author, and price. Implement:

- A. A default constructor that initializes all attributes to default values ("Unknown", "Unknown", 0.0).
 - B. A parameterized constructor that accepts values for all three attributes.
 - C. A display() member function to print the book details.
- In main(), create one object using each constructor and display its details.

Your Code

```
1 #include<iostream>
2 using namespace std;
3 class book{
4     string title,author;
5     float price;
6 public:
7     book() {
8         title="unknown";
9         author="unknown";
10        price=0.0;
11    }
12    book(string t,string a,float p)
13    {
14        title=t;
15        author=a;
16        price=p;
17    }
18    void display() {
19        cout<<"Title: "<<title<<endl;
20        cout<<"Author: "<<author<<endl;
21        cout<<"Price: "<<price<<endl;
22    }
23 };
24 int main() {
```

Input

```
God's Nature
Mohan Rao
500
```

Output



Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Q2 Constructor Overloading

20 marks

Write a C++ program that defines a class Rectangle with the following constructors:

- A. A default constructor that sets length and breadth to 1.
- B. A single-parameter constructor that creates a square (length = breadth).
- C. A two-parameter constructor that sets both length and breadth.

Include a member function area() to compute the rectangle's area. In main(), create three objects using each constructor and print their areas.

Your Code

```
1 #include<iostream>
2 using namespace std;
3 class Rectangle{
4     float length,breadth;
5 public:
6     Rectangle(){
7         length=1;
8         breadth=1;
9     }
10    Rectangle(float side){
11        length =side;
12        breadth =side;
13    }
14    Rectangle(float l,float b){
15        length =l;
16        breadth=b;
17    }
18    float area(){
19        return length *breadth;
20    }
21 };
22 int main(){
23     float side,length,breadth;
24     cin>>side;
25     cin>>length>>breadth;
26     Rectangle r1;
```

Input

```
5
10 4
```

Output



Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Q3 Copy Constructor with Dynamic Memory

20 marks

Write a C++ program that defines a class Student containing a dynamically allocated character array for the student's name and an integer for marks. Implement:

- A. A parameterized constructor that allocates memory and copies the name.
- B. A deep copy constructor that allocates new memory for the copied object.
- C. A destructor that releases the allocated memory.
- D. A display() function to print the student's details.

In main(), create a Student object, create another object using the copy constructor, and modify the second object's name to demonstrate that the original remains unaffected.

Your Code

```
1 #include<iostream>
2 #include<cstring>
3 using namespace std;
4 class Student{
5     char*name;
6     int marks;
7     public:
8     Student(const char n[],int m){
9         name = new char[strlen(n) + 1];
10        strcpy(name,n);
11        marks = m;
12    }
13    Student(const Student &s){
14        name=new
15        char[strlen(s.name) + 1];
16        strcpy(name,s.name);
17        marks=s.marks;
18    }
19    ~Student(){
20        delete[] name;
21    }
22    void display(){
23        cout<<"Name: "<<name<<endl;
24        cout<<"Marks: "<<marks<<endl;
25    }
26    void changename(const char *n){
27        strcpy(name , n);
```

Input

Pavan 80
Ravi

Output



Visible Test Cases

Test Case 1	☐
Test Case 2	☐
Test Case 3	☐

Q4 Destructor and Resource Management

20 marks

Write a C++ program that defines a class FileHandler which simulates opening and closing a file. Implement:

A. A constructor that prints "File <name> opened".

B. A destructor that prints "File <name> closed".

In main(), create three FileHandler objects inside a block {} and observe the order in which constructors and destructors are invoked. Print a message after the block ends to confirm the destruction order.

Your Code

```
1 #include<iostream>
2 using namespace std;
3 class FileHandler{
4     char name;
5     public:
6     FileHandler(char n){
7         name=n;
8         cout<<"file "<<name<<" opened"<<endl;
9     }
10    ~FileHandler(){
11        cout<<"file "<<name<<"closed"<<endl;
12    }
13 };
14 int main(){
15     char a, b, c;
16     cin >> a >> b >> c;
17     {
18         FileHandler f1(a);
19         FileHandler f2(b);
20         FileHandler f3(c);
21     }
22     cout<<"Block ended"<<endl;
23     return 0;
24 }
```

Input

A
B
C

Output



Visible Test Cases

Test Case 1	☐
Test Case 2	☐
Test Case 3	☐

Q5 Overloading the + Operator (Complex Numbers)

20 marks

Write a C++ program that defines a class Complex with real and imaginary parts. Overload the + operator to add two complex numbers and return a new Complex object. Include a display() function to print the result in the form a + bi. In main(), create two Complex objects, add them using the overloaded + operator, and display the result.

Your Code

```
1 #include<iostream>
2 using namespace std;
3 class complex {
4     public:
5         int real,imag;
6         complex(int r,int i){
7             real=r;
8             imag=i;
9         }
10        complex operator + (complex c){
11            return complex(real + c.real,imag + c.imag);
12        }
13        void display(){
14            cout<<real<<" + "<<imag<<"i";
15        }
16 };
17 int main(){
18     int a,b,c,d;
19     cin>>a>>b;
20     cin>>c>>d;
21     complex n1(a,b);
22     complex n2(c,d);
23     complex result=n1 +n2;
24     result.display();
25     return 0;
26 }
```

Input

```
3 4
5 2
```

Output**Visible Test Cases**

Test Case 1

Test Case 2

Test Case 3